AI Lab
Week 2

Solve 8 puzzle problems

Implement Iterative deepening search algorithm

8 Puzzle Problem

5	4	
6	1	8
7	3	2

Initial State

	1	2
3	4	5
6	7	8

Goal State

- The 8-puzzle consists of a 3×3 board with eight numbered tiles and a blank space.
- A tile adjacent to the blank space can slide into the space.
- The object is to reach a specified goal state.

Solve 8 puzzle problems

- First Solve 8- Puzzle problem using Depth First Search (DFS)

Implement Iterative deepening search (IDS) algorithm

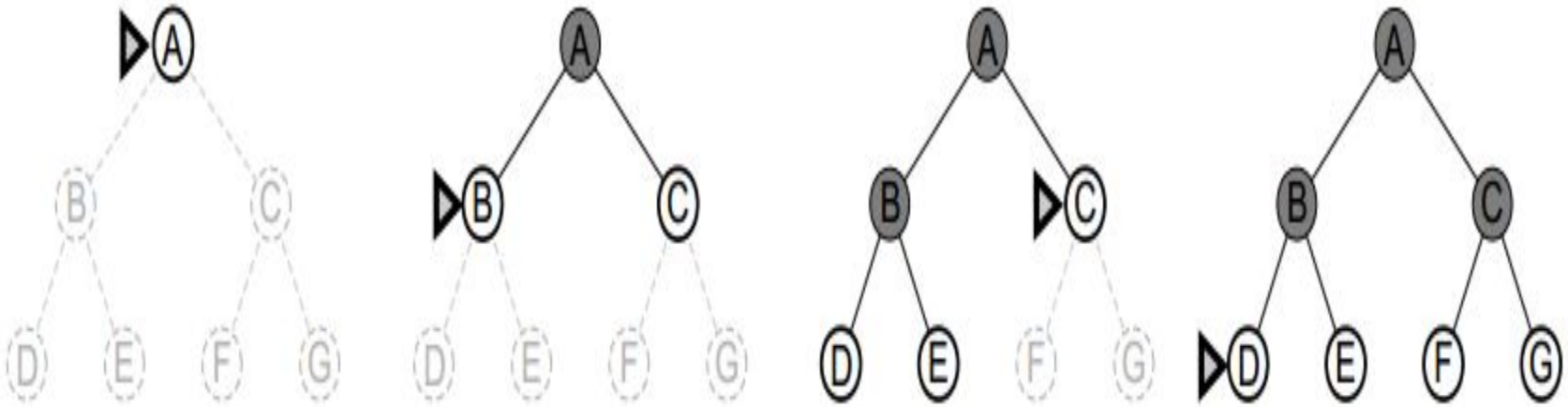
Solve 8-Puzzle Problem using IDS algorithm

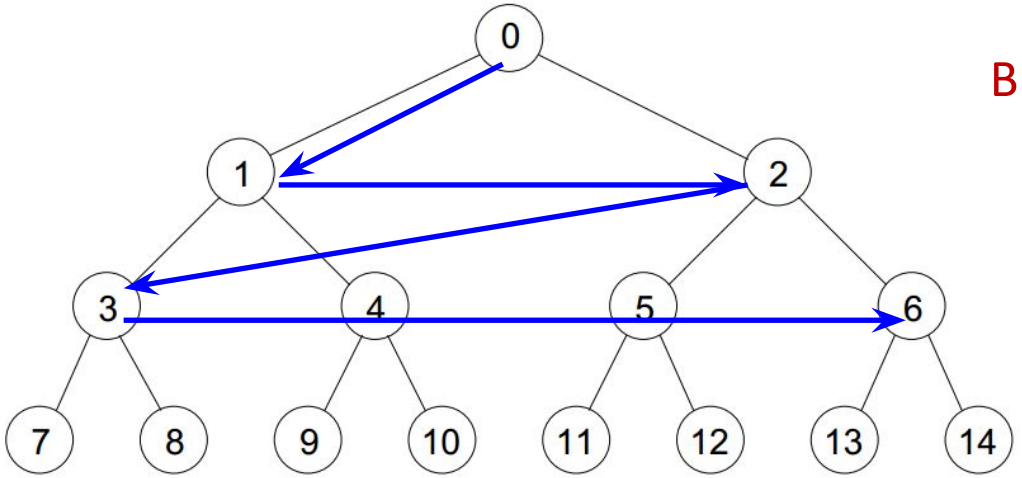
Depth First Search

DFS: We traverse all adjacent one by one, when we traverse an adjacent, we finish traversal of all vertices reachable through the adjacent completely. After we finish one adjacent and its reachable, we go to the next adjacent and finish all reachable through next and continue this way. Similar to tree where we first completely traverse the left subtree and then go to the right subtree.

Breadth-First Search

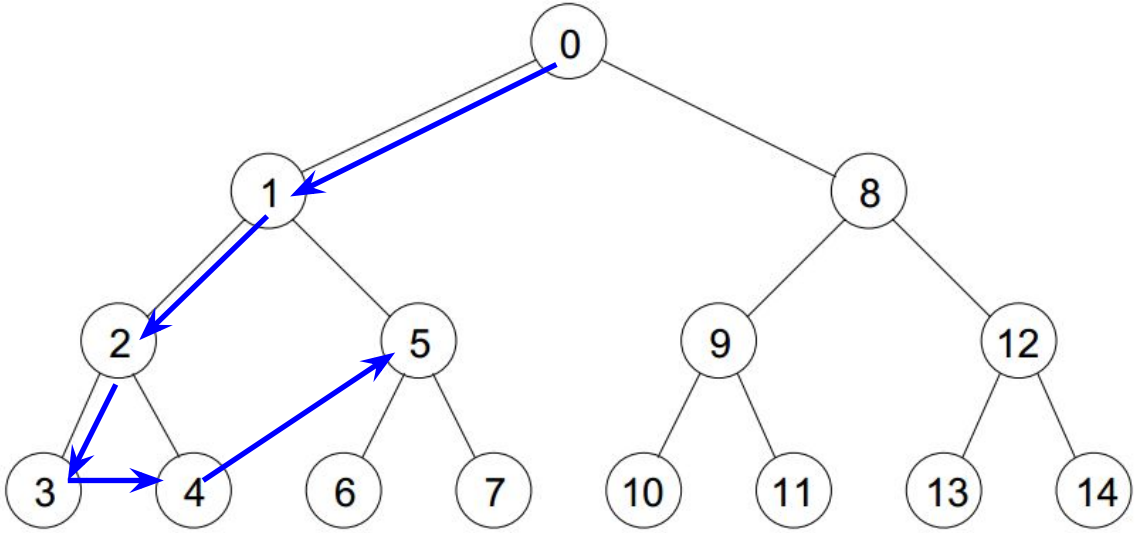
- BFS: It begins with a node, then first traverses all its adjacent. Once all adjacent are visited, then their adjacent are traversed.





Breadth First Search

Depth First Search

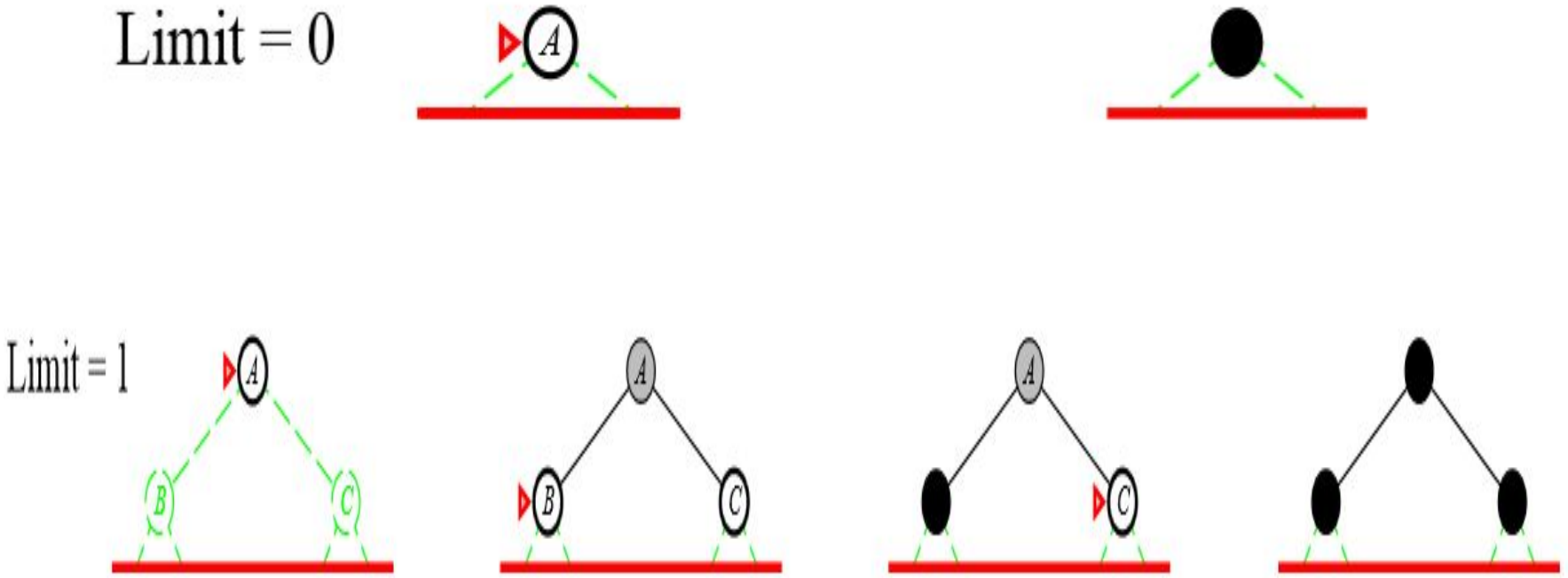


Iterative Deepening Search(IDS) or Iterative Deepening Depth First Search(IDDFS)

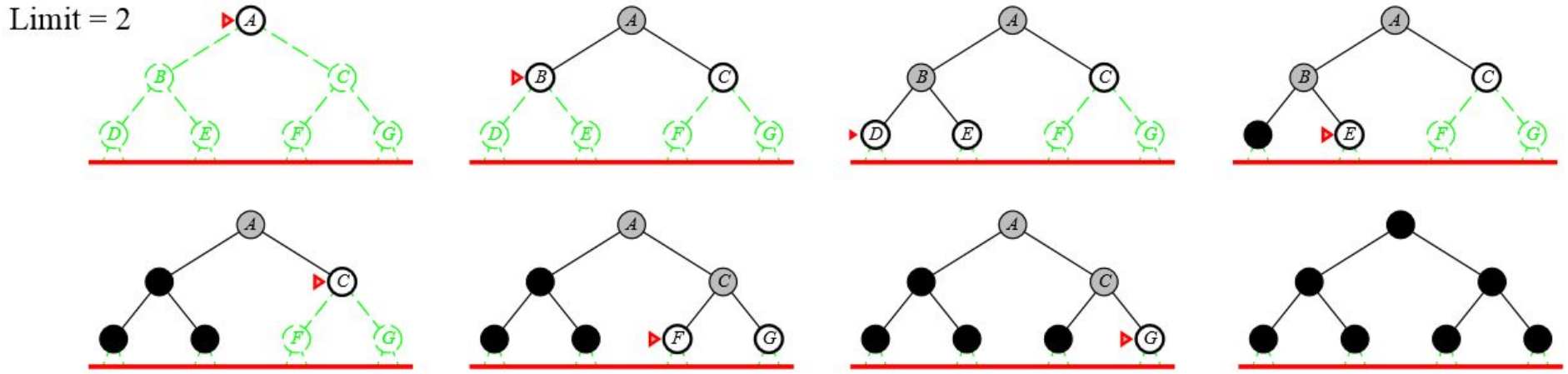
- IDDFS calls DFS for different depths starting from an initial value. In every call, DFS is restricted from going beyond given depth. So basically we do DFS in a BFS fashion.

```
function ITERATIVE-DEEPENING-SEARCH( problem) returns a solution
    inputs: problem, a problem
    for depth ← 0 to ∞ do
        result ← DEPTH-LIMITED-SEARCH( problem, depth)
        if result ≠ cutoff then return result
    end
```

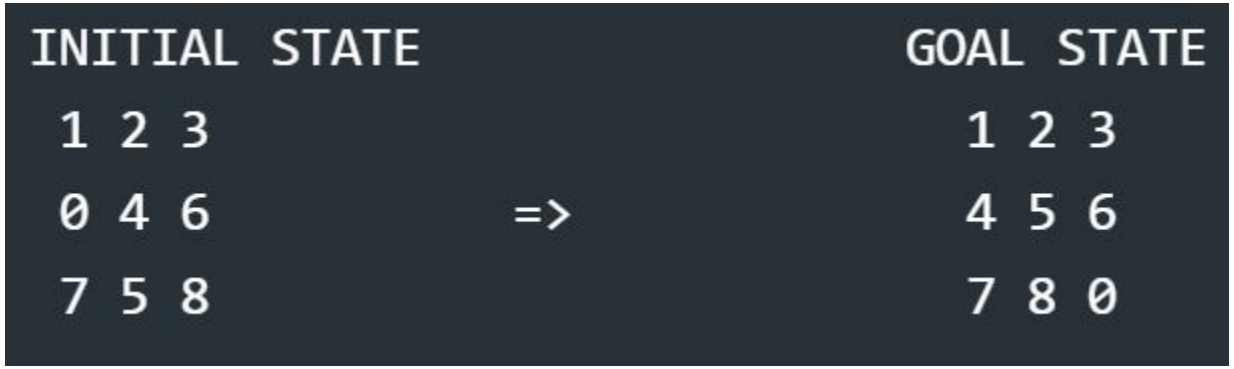
Iterative deepening search (or iterative deepening depth-first search)

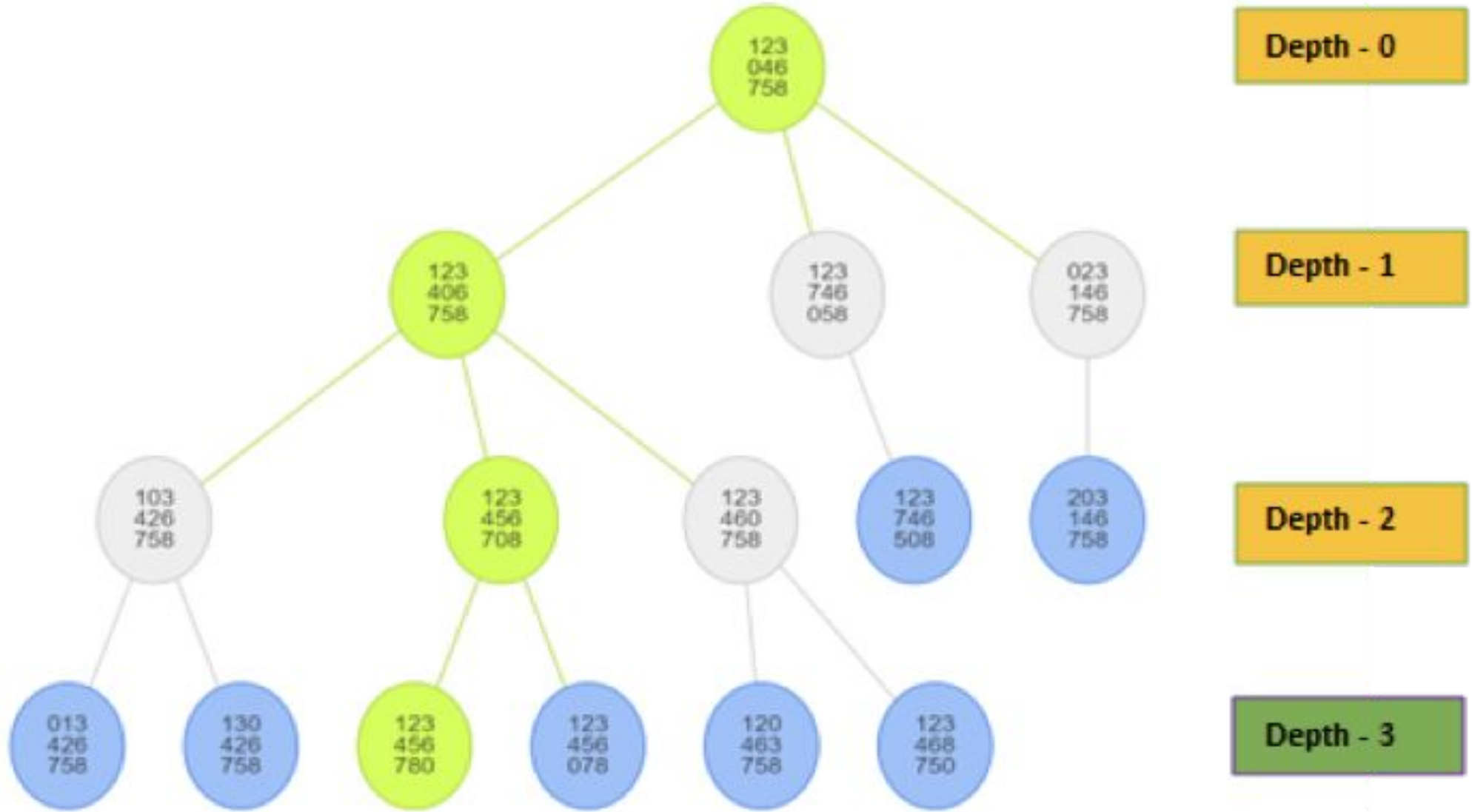


Iterative Deepening Search(IDS) or Iterative Deepening Depth First Search(IDDFS)



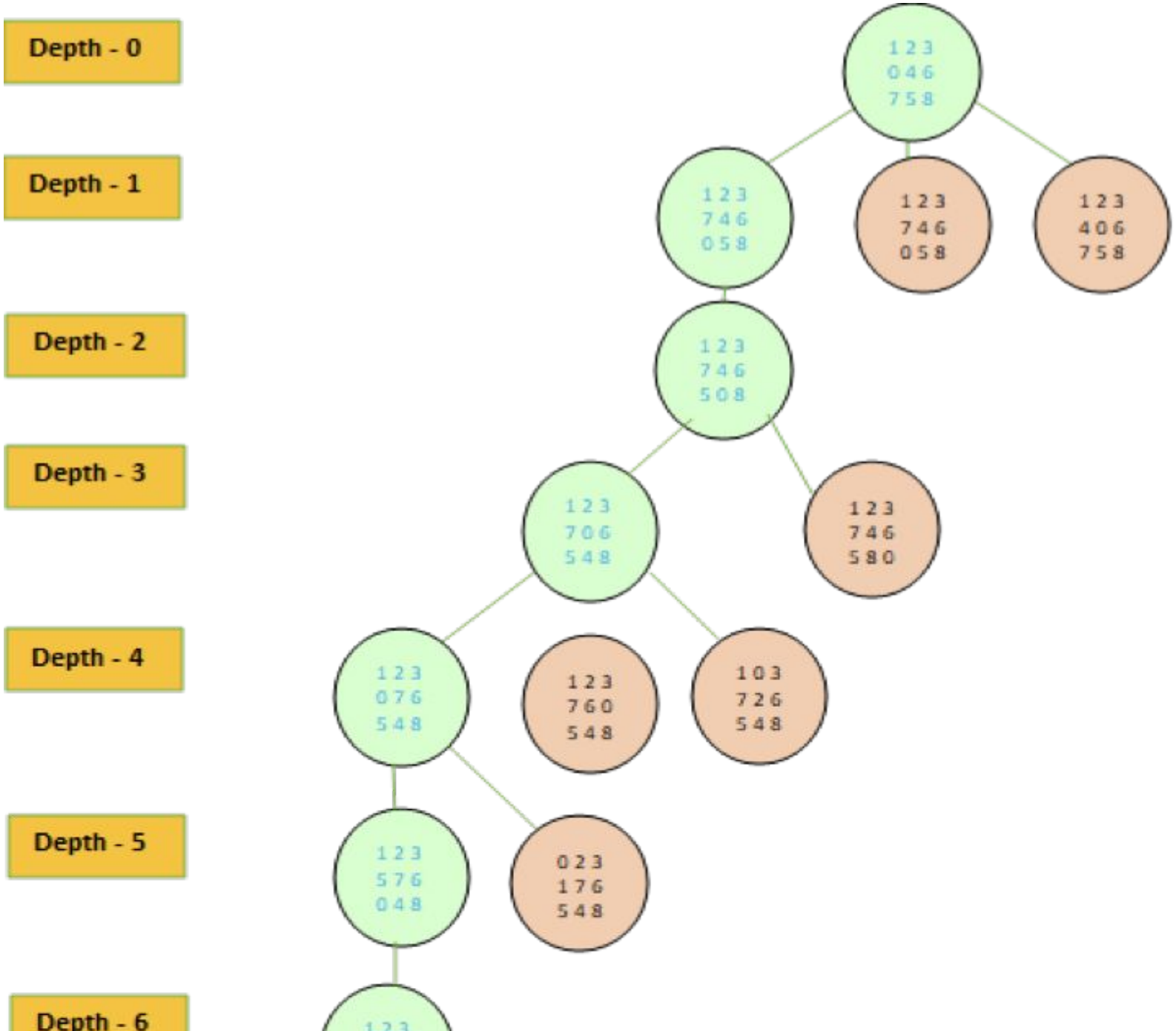
8 Puzzle Problem



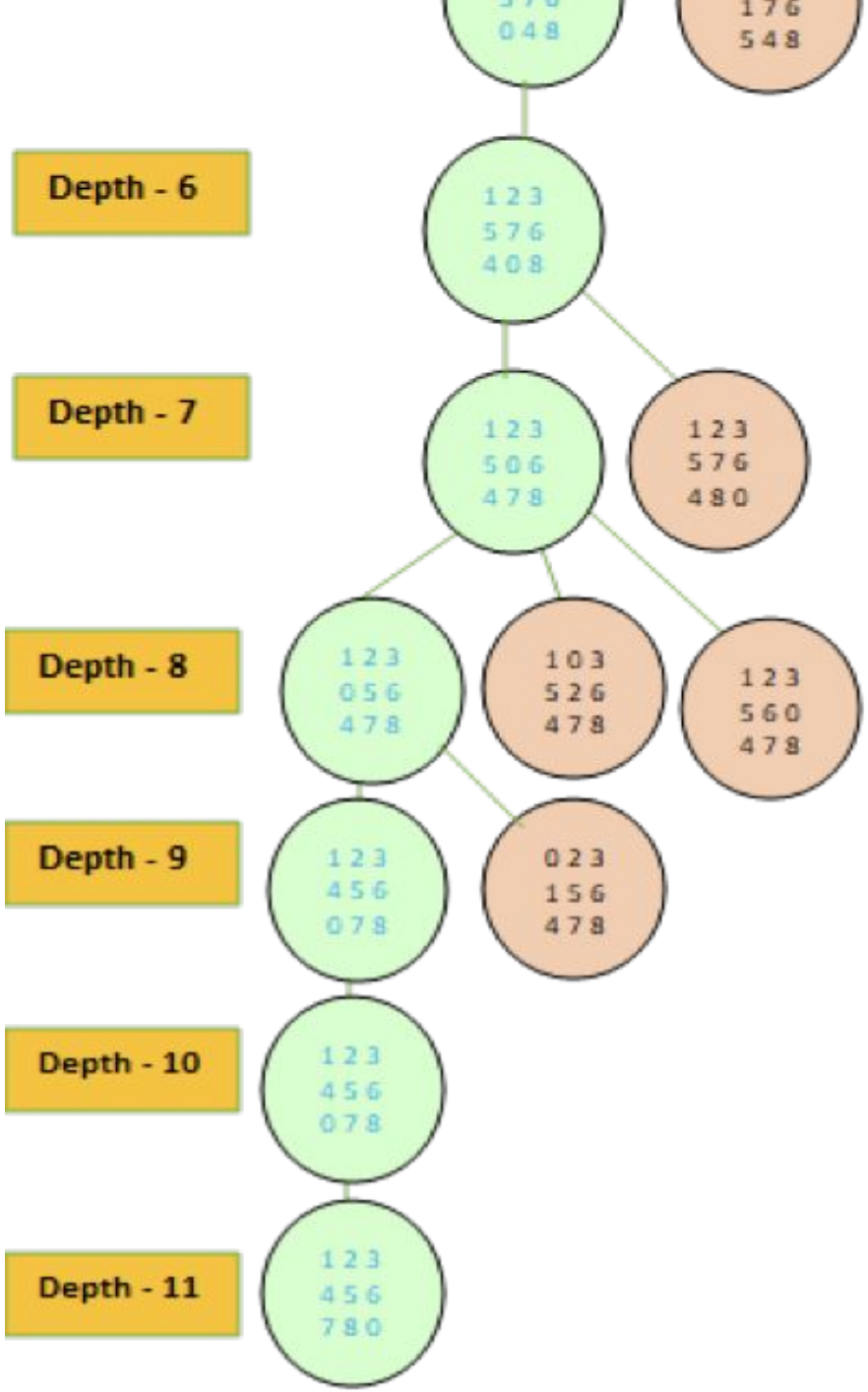


BFS Tree Example (We have found out our result in DEPTH 3 Only)

DFS


Depth - 0
Depth - 1
Depth - 2
Depth - 3
Depth - 4
Depth - 5
Depth - 6
1 2 3
0 4 6
7 5 8
1 2 3
7 4 6
0 5 8
1 2 3
7 4 6
0 5 8
1 2 3
4 0 6
7 5 8
1 2 3
7 4 6
5 0 8
1 2 3
7 0 6
5 4 8
1 2 3
7 4 6
5 8 0
1 2 3
0 7 6
5 4 8
1 2 3
7 6 0
5 4 8
1 0 3
7 2 6
5 4 8
1 2 3
5 7 6
0 4 8
0 2 3
1 7 6
5 4 8

DFS


048
176
548
Depth - 6
123
576
408
Depth - 7
123
506
478
123
576
480
Depth - 8
123
056
478
103
526
478
123
560
478
Depth - 9
123
456
078
023
156
478
Depth - 10
123
456
078
Depth - 11
123
456
780